Advanced Computer Networks

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MAC Protocol: ALOHA Tutorial

DSSRG: Decentralized Smart Systems Research Group

https://sites.google.com/iitbbs.ac.in/dssrg
Or Google dssrg iitbbs

Basic goals

- ALOHA

Question 1

Consider a Pure ALOHA network transmitting 200-bit frames over a 200 kbps shared channel. Determine the condition that must be met to prevent frame collisions.

Question 1

- Frame size = 200 bits
- Channel speed = 200 kbps (kilobits per second)

  = 200,000 bits per second

Time to transmit one frame:

Transmission time=Frame size/Data rate

=200 bits/200,000 bits/sec

=0.001 seconds

=1 millisecond (ms)

Question 1

In Pure ALOHA, the vulnerable time = 2×Transmission time

Why?

Because a frame can collide if:

- Another frame starts **up to 1 transmission time before**, or
- Another frame starts **up to 1 transmission time after**

So total vulnerable time:

$$2 \times 1\text{ms} = 2\text{ms}$$

To make it collision-free, no other station should transmit within 2 ms of any transmission.

Question 2

A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces 1000 frames per second (including retransmitted frames)?

Question 2

Frame size = **200 bits**

Channel speed = **200 kbps = 200,000 bits/sec**

Frames generated = **1000 frames per second**

Transmission time = Frame size/Data rate

= 200/200,000

= 0.001 seconds

= 1 ms

Question 2

G = Frames per second × Transmission time

=> G = 1000×0.001

=> G = 1

So, offered load G = 1

Apply Pure ALOHA Formula:

$$S = Ge^{-2G}$$

$$\Rightarrow S = 1 \times e^{-2}$$

$$\Rightarrow S \approx 0.135 \text{ frames per unit time}$$

$$\Rightarrow S = 0.135/0.001 \text{ frames/sec}$$

$$\Rightarrow S = 135 \text{ frames/sec}$$

Question 3

A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput if the system (all stations together) produces 1000 frames per second (including retransmitted frames).

Question 3

Frame size = **200 bits**

Channel speed = **200 kbps = 200,000 bits/sec**

Frames generated = **1000 frames per second**

Transmission time = Frame size/Data rate

= 200/200,000

= 0.001 seconds

= 1 ms

Question 3

G = Frames per second × Transmission time

=> G = 1000×0.001

=> G = 1

So, offered load G = 1

Apply Slotted ALOHA Formula:

$$S = Ge^{-G}$$

$$\Rightarrow S = 1 \times e^{-1}$$

=> S ≈ 0.368 frames per unit time

=> S = 0.368/0.001 frames/sec

=> S = 368 frames/sec

Question 4

A group of N stations shares 50 Kbps pure ALOHA channel. Each station outputs a 500-bit frame on an average of once 5000 ms, even if the previous one has not been sent. What is the maximum value of N?

Question 4

Channel rate = **50 kbps = 50,000 bits/sec**

Frame size = **500 bits**

Each station generates 1 frame every **5000 ms or 5 seconds**

Frame Transmission Time = Frame size/ Data rate

= 500/ 50,000

= 0.01 seconds

= 10 ms

Question 4

Maximum Throughput of Pure ALOHA = 0.184

Maximum usable channel capacity = 0.184×50,000

= 9,200 bits/sec

= 18.4 frames/sec

Each station generates ⅕ = 0.2 frames/sec

So total offered load = 0.2 * N frames/sec

For maximum N, 0.2 * N = 18.4

=> N = 92